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Education

Massachusetts Institute of Technology

Cambridge, MA

PH.D. CANDIDATE IN PHYSICS

2014 - Present

- Research in Deep Learning and Applied Physics.
- Coursework in CS and ML: Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Statistical Learning Theory, Computer Vision, Natural Language Processing

Peking University

Beijing, China

B.S. IN PHYSICS, B.A. IN ECONOMICS (DUAL MAJOR)

2010 - 2014

Research in Quantum Information and Quantum Computation

Selected Award

2010 **Gold Medal**, 41th International Physics Olympiad (IPhO)

Zagreb, Croatia

Experience

Deep Learning and Applied Physics Group in MIT

Cambridge, MA

RESEARCH ASSISTANCE, ADVISOR: MARIN SOLJAČIĆ

2016 - Present

- Applied a variety of machine learning methods to physics problems, e.g. photonics inverse design.
- Optimized architecture of optical neural network chip; developed hardware level training algorithms.
- Developed efficient unitary recurrent neural network, published on ICML.
- Supervised several graduate and undergraduate students researching in varieties of topics including RNN, GAN, Reinforcement Learning etc.

Center for Ultracold Atoms in MIT

Cambridge, MA

RESEARCH ASSISTANCE, ADVISOR: WOLFGANG KETTERLE

2014 - 2016

- Built high power laser tapered amplifier; designed analog PID circuits used to generally stabilize laser power.
- Developed atomic physics image software system for AMO physics.

Quantum Information Group in Chinese Academy of Sciences

Beijing, China

UNDERGRADUATE RESEARCHER, ADVISOR: HENG FAN

2011 - 2014

- Investigated optimization problems in quantum cloning and quantum cryptography.

Qidian Technology

Beijing, China

QUANTITATIVE RESEARCHER AND TRADOR

2012 - 2014

- Collected massive data from public social media website including Linkedin, Twitter, Facebook and Weibo.
- Built natural language models to evaluate employee performances in NASDAQ high-tech companies based on social media data.
- Established financial models using deep neural network to predict company performances given employee performances.
- Achieved 20% per year profit rate

Selected Publications

2017	Gated Orthogonal Recurrent Units: On Learning to Forget , Li Jing, Caglar Gulcehre, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljačić, Yoshua Bengio	AAAI under review
2017	Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs , Li Jing, Yichen Shen, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljačić	ICML
2014	Quantum Cloning Machines and the Applications , Heng Fan, Yi-Nan Wang, Li Jing, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu	Phys. Rep.

Skills

Language	English, Mandarin
Deep Learning Frameworks	Tensorflow, PyTorch, Theano
Programming	Python, C++, Matlab, Lua
Full Stack Developer	Javascript, Django, AngularJS, NodeJS, Less, PHP, Swift.